THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 1: **Exploring Substances: Acidic, Basic and Neutral** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A student mixes equal volumes of acid and base. The mixture still turns blue litmus red. This suggests: 1

(a) the acid was stronger or more concentrated
(b) the base was stronger
(c) the mixture is neutral
(d) no reaction

2. A cook adds a pinch of baking soda to lemon-flavoured drink and it fizzes. This shows: 1

(a) salt reaction **(b)** acid-carbonate reaction
(c) no reaction **(d)** sugar reaction

3. Which of the following reaction shows a salt being formed? 1

(a) $2\,Mg + O_2 \longrightarrow 2\,MgO$ **(b)** $C + O_2 \longrightarrow CO_2$ **(c)** $2\,H_2 + O_2 \longrightarrow 2\,H_2O$ **(d)** $HCl + NaOH \longrightarrow NaCl + H_2O$

4. An indicator is green in a solution. A drop of unknown liquid Z turns it red. Z is: 1

(a) basic **(b)** neutral **(c)** salt **(d)** acidic

5. Which one of the following is both acidic in taste and used as a food preservative? 1

(a) lime water **(b)** soap **(c)** baking soda **(d)** vinegar

6. In which of the following will the red litmus turn blue? 1

(a) lemon **(b)** vinegar **(c)** aqueous $NaOH$ **(d)** aqueous HCl

7. Which of these will not show a colour change in turmeric? 1

(a) soap solution **(b)** washing soda **(c)** sugar solution **(d)** lime water

8. Which of these household items would best relieve the pain of a sting that contains formic acid? 1

(a) tamarind pulp **(b)** vinegar
(c) a paste of baking soda and water **(d)** lemon juice

9. **Assertion (A):** Neutralisation is the reaction between an acid and a base to form salt and water. 1
Reason (R): Heat is released during a neutralisation reaction.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Antacid tablets give relief from indigestion. 1
Reason (R): Antacids are bases that neutralise the excess acid in the stomach.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why is soap solution slippery to touch? 2

12. What could be used to neutralise a spilled acid in the lab safely? 2

13. Why is a concentrated acid kept in a bottle with a clear label and a tight cap? 2

14. Why must acidic waste from factories not be released into rivers? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how baking soda in a cake batter makes the cake rise. 3

16. A factory discharges waste that turns blue litmus red. Suggest a treatment method. 3

17. Identify the odd one: vinegar, lemon, baking soda, curd. Give a reason. 3

OR

A person's teeth show decay. How are acids involved and what preventive steps are advised? 3

18. Explain how a baking soda solution can help stop the pain of an ant bite and how to check if the area is neutral. 3

OR

Explain what happens to turmeric stain on a white shirt when soap is applied and then lemon juice is added. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** In the school laboratory, Meena crushes red China rose petals in water and filters the liquid to make an indicator. She adds a few drops to three samples: lemon juice, soap solution and sugar solution. 5
(i) What colour does the China rose indicator give with lemon juice?
(ii) What colour does it give with soap solution?
(iii) What happens in the sugar solution?
(iv) A fourth liquid turns turmeric paper red. State its nature and what it will do to red litmus paper, giving a reason.

20. A test for acid rain is planned for your town. Describe steps, tools and conclusions. 5

OR

A student suspects that a drinking water sample is acidic. Plan the test and a safe next step. 5

END OF QUESTION PAPER — Total: 40 Marks